

Analysis of finite hypothesis class

Setup

Sometimes also union bound version

$h_S \in \text{ERM}_H(S)$, assume $|\mathcal{H}| < \infty$,

assume realizability ($\exists h^* \in \mathcal{H}$, $L_{D,f}(h^*) = 0$)

and have binary classification w/ 0-1 loss

$$\text{so } L_{D,f}(h) = \mathbb{P}_{x \sim D} (h(x) \neq f(x))$$

Goal is PAC learning: $\forall \epsilon, \delta \in (0, 1)$

prove (if m sufficiently large)
 $L_{D,f}(h_S) \leq \epsilon$ w.p. $\geq 1 - \delta$

$$\text{aka } \mathcal{D}^m(\{S : L_{D,f}(h_S) > \epsilon\}) < \delta$$

Analysis: Fix $\epsilon > 0$

let $\mathcal{H}_B$ = "bad hypotheses"

$\mathcal{H}_B \neq \mathcal{H}$ since $L_{D,f}(h^*) = 0$
by assumption

Fix $m \in \mathbb{N}$

$$= \{h \in \mathcal{H} : L_{D,f}(h) > \epsilon\} \subseteq \mathcal{H}$$

so $|\mathcal{H}_B| \leq |\mathcal{H}|$ cardinality, not abs. value

let M = Mistake-making samples

$$= \{S : \exists h \in \mathcal{H}_B \text{ s.t. } \underbrace{h \in \text{ERM}_H(S)}_{\text{equiv. } \hat{L}_S(h) = 0} \}$$

equiv., $\hat{L}_S(h) = 0$
by realizability

Our $h_S \in \text{ERM}_H(S)$

so if $S \notin M$ we know h_S is good.

How big is M ?

Idea: if h is bad, chance

$$M = \bigcup_{h \in \mathcal{H}_B} \{S : \hat{L}_S(h) = 0\}$$

$\hat{L}_S(h) = 0$ is small

$$\text{so } \mathcal{D}^m(M) \leq \sum_{h \in \mathcal{H}_B} \mathcal{D}^m(\{S : \hat{L}_S(h) = 0\})$$

via union bound

P. 1

(aside: union bound: $D(A \cup B) \leq D(A) + D(B)$
 $\uparrow$ "or"

Directly useful for bounding failures.

if true risk is 0, empirical risk is 0

For successes, use $D(A) = 1 - D(A^c)$

and De Morgan's laws: $(\cup A_i)^c = \cap A_i^c$

$$(\cap A_i)^c = \cup A_i^c$$

Each term in sum: $D^m(\{S: \hat{L}_S(h) = 0\})$, fix $h \in \mathcal{H}_B$

TRICK!

Now independent since not random

if true risk is large, it's unlikely empirical risk is small

Since $\hat{L}_S(h) = 0$ means $h(x_i) = f(x_i) \forall i \in [m]$
 $\underbrace{\hspace{1cm}}$ true label

$$D^m(\dots) = D^m(\{S: h(x_i) = f(x_i) \forall i \in [m]\})$$

$$= \prod_{i=1}^m D(\{x: h(x) = f(x)\}) \text{ by independence}$$

$$= D(\{x: h(x) = f(x)\})^m \text{ since identically distr.}$$

$$= (1 - D(\{x: h(x) \neq f(x)\}))^m \text{ take complement}$$

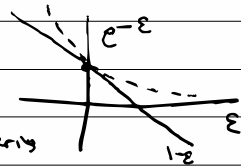
$$= (1 - \underbrace{L_{D,f}(h)}_{> \epsilon})^m \text{ since } 0-1 \text{ loss}$$

$$\leq (1 - \epsilon)^m$$

$$\leq e^{-\epsilon \cdot m} \text{ since } (1 - \epsilon) \leq e^{-\epsilon}$$

nicer looking

proof via Taylor Series or derivative + convexity



Going back to union bound,

$$D^m(M) \leq \sum_{h \in \mathcal{H}_B} e^{-\epsilon m} = |\mathcal{H}_B| \cdot e^{-\epsilon m} \leq \underbrace{|\mathcal{H}|}_{\text{"}f\text{"}} \cdot e^{-\epsilon m}$$

Want $\delta = |\mathcal{H}| \cdot e^{-\epsilon m}$, or, $-\epsilon m = \log(\delta/|\mathcal{H}|)$

i.e. $m = \frac{1}{\epsilon} \log\left(\frac{|\mathcal{H}|}{\delta}\right)$

So...

Corollary 2.3 (binary class, 0-1 loss, realizable, $|\mathcal{H}| < \infty$)
 $\forall \delta \in (0, 1) \forall \epsilon > 0$, if $m \in \mathbb{Z}$ satisfies

$$m \geq \frac{1}{\epsilon} \log\left(\frac{|\mathcal{H}|}{\delta}\right) \text{ then } \forall f, \forall D$$

(as long as realizable), if S has m iid samples
 then $\forall h_S \in \text{ERM}_{\mathcal{H}}(S)$, w. prob $\geq 1 - \delta$,

$$\underbrace{L_{D,f}(h_S)}_{\text{"approximate"}} \leq \underbrace{\epsilon}_{\text{"probably"}}$$

m is our "sample complexity":

smaller δ
 and/or
 smaller $\epsilon \Rightarrow$ need more samples

Later we'll drop realizability assumption and
 allow for "agnostic" case, and also more
 general loss functions: Corollary 4.6

Then $m \geq \frac{1}{\epsilon^2} \log\left(\frac{2|\mathcal{H}|}{\delta}\right)$
 $\uparrow$ yuck!

$\log(1/\delta)$
 common

$\frac{1}{\epsilon^2}$ nice,

$\frac{1}{\epsilon^2}$ more common

So, from training examples, we can learn arbitrarily
 well (given enough data)

... if our choice of $\mathcal{H}$ (inductive bias) was good enough to include a good classifier. } Assumption / inductive bias

more generally, (§3), define

Def A hypothesis class $\mathcal{H}$ is "PAC learnable"

if (ϵ, δ)

$\exists$ function $m_{\mathcal{H}} : (0, 1)^2 \rightarrow \mathbb{N}$ "sample complexity"

and some learning algo "A" (eg. ERM $_{\mathcal{H}}$)

such that

algo A

must work
for all distributions

$\forall D$ over X , $\forall f: X \rightarrow \{0, 1\}$

→ still doing binary
classif.

if $(\mathcal{H}, D, f)$ is realizable, then

if $m \geq m_{\mathcal{H}}(\epsilon, \delta)$ iid samples are used in algo A,

A returns a classifier/hypothesis h s.t. w.p.

$\geq 1 - \delta$, $L_{D, f}(h) \leq \epsilon$.

Agnostic PAC learning means relaxing assumption of an oracle $f: X \rightarrow Y$ i.e. no longer need be consistent

Now, let D be a ^{joint} distribution over $Z := X \times Y$
and D_x its marginal wrt X ,
 $D(Z/x)$ its conditional.

Before, in (pure) PAC learning, we're assuming

$D(Z/x)$ collapses to a deterministic function $f(x)$

$\hat{L}_g$ is unchanged but (for 0-1 loss)

$$\text{turn } L_{D,f}(h) = \mathbb{P}_{x \sim D} (h(x) \neq f(x))$$

$$\text{into } L_D(h) = \mathbb{P}_{(x,y) \sim D} (h(x) \neq y) \quad \left. \vphantom{L_D(h)} \right\} \text{AGNOSTIC CASE}$$

Now, it's typically too much to ask for $L_D(h) \leq \epsilon$

...

△ One algo for all distributions

Def H is agnostic PAC learnable if $\exists m_H: (0,1)^2 \rightarrow \mathbb{N}$,
 $\exists$ algo A , s.t. $\forall \epsilon, \delta \in (0,1)$, $\forall$ distr. D over $X \times Y$,
 A s.t. if S has $m \geq m_H(\epsilon, \delta)$ iid samples, then
w.p. $\geq 1 - \delta$
$$L_D(h) \leq \epsilon + \underbrace{\min_{h' \in H} L_D(h')}_{\text{new}}$$

where $h = ACS$

Note: Mohri puts it this way: agnostic = inconsistent + stochastic
consistent: means $\min_{h' \in H} L(h') = 0$
deterministic: means $D(y(x) = f(x))$

Remark An alternative comparison would be the Bayes Optimal Predictor, $f_D(x) = \begin{cases} 1 & \text{if } P(y=1/x) \geq 1/2 \\ 0 & \text{else} \end{cases}$ } if binary classif.
uses knowledge of D !

Optimal, but

- ① you have to know D
- ① it changes when D changes
- ② it's asking too much

Loss function $\hat{L}_S(h) = \frac{1}{n} \sum_{i=1}^n l(h, z_i)$
Allow general risk/loss $L_D(h) = \mathbb{E}_{z \sim D} l(h, z)$

Ex: 0-1 loss for binary classif,

$$l_{0-1}(h, (x, y)) = \begin{cases} 0 & h(x) = y \\ 1 & \text{else} \end{cases} = \mathbb{I}_{h(x) \neq y}(h, z)$$

Ex regression, l_p losses

$$l_p(h, (x, y)) = (h(x) - y)^p, \quad p \geq 1 \text{ for convex}$$

$p=2$ most common $p=1$

Brief History:

a lot of the framework due to
Vapnik + Chervonenkis '71
PAC due to Valiant '84

Vladimir Vapnik (1936-) from Samarkand (Uzb.) PhD '64

'71 VC theory

'90 move to US, AT&T, develop SVMs

'98 "Stat. Learning Theory" book 60,000+ citations

'02 NEC

100,000+

'03 Columbia

'14 Facebook AI Research, '16 Venere (abs

Alexey Chervonenkis (1938-2014)

'71 VC theory

Kept researching, stayed in USSR/Russia

Leslie Valiant (1949-) from Budapest

Prof. at Harvard since '82

Many works in Theoretical Computer Science (TCS)

two sons on faculty at Stanford + Brown

5b: detour, convergence of functions

Thursday, May 14, 2026

2:07 PM

Detour: convergence of functions

Added '26

Functions $f: \Omega \rightarrow \mathbb{R}$

① Def Pointwise convergence $f_n \xrightarrow{pw} f$ means $\forall x \in \Omega, f_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$
i.e.

$$\underline{\forall x \in \Omega}, \forall \varepsilon > 0, \exists N \text{ s.t. } n \geq N \Rightarrow |f_n(x) - f(x)| < \varepsilon$$

② Def Uniform convergence $f_n \xrightarrow{u} f$ or $\|f_n - f\|_{\infty} \rightarrow 0$ or $\|f_n - f\|_{\infty} \rightarrow 0$
means

$$\forall \varepsilon > 0, \exists N \text{ s.t. } n \geq N \Rightarrow \underbrace{\forall x \in \Omega, |f_n(x) - f(x)| < \varepsilon}_{\text{equiv. } \|f_n - f\|_{\infty} < \varepsilon}$$

③ Def L^2 convergence $f_n \xrightarrow{L^2} f$ or $\|f_n - f\|_{L^2} \rightarrow 0$
means

$$\forall \varepsilon > 0, \exists N \text{ s.t. } n \geq N \Rightarrow \|f_n - f\|_2 < \varepsilon$$
$$:= \sqrt{\int |f_n(x) - f(x)|^2 dx}$$

EXERCISE: for each seq. of fcn, decide/prove
which types of convergence it has

Trick: find limit f first.

A) $f_n(x) = x^n$ on $[0, 1]$

B) $f_n(x) = x^n$ on $[0, 1)$

C) $f_n(x) = x^2 + \frac{1}{n} \sin(x)$ on $[0, 2\pi]$

D) typewriter sequence on $[0, 1]$

$f_1 = \begin{array}{c} \text{---} \\ 0 \quad 1 \end{array}$ $f_2 = \begin{array}{c} \text{---} \\ 0 \quad .5 \end{array}$ $f_3 = \begin{array}{c} \text{---} \\ 0 \quad .5 \quad 1 \end{array}$

$f_4 = \begin{array}{c} \text{---} \\ 0 \quad .25 \end{array}$ $f_5 = \begin{array}{c} \text{---} \\ .25 \quad .5 \end{array}$... $f_7 = \begin{array}{c} \text{---} \\ .75 \quad 1 \end{array}$

$f_8 = \begin{array}{c} \text{---} \\ 0 \quad .125 \end{array}$... $f_{15} = \begin{array}{c} \text{---} \\ .75 \quad 1 \end{array}$
etc

5c: detour, convergence of functions

Thursday, May 14, 2026

2:17 PM

Do after discussing convergence of r.v.

In 2026, that was after ch. 14 SGD!

Detour: random variables

In our PAC learning, S is a random variable, hence so is

$$\theta_m := L_{D,f}(A(S)), \quad \theta_m \geq 0$$

We're proving things like $\theta_m \leq \varepsilon$ w.p. $\geq 1 - \delta$ (if $m \geq m(\varepsilon, \delta)$)

How does this relate to our usual notions of convergence of r.v.?

You can show existence of finite $m(\varepsilon, \delta)$ is equivalent

$$\text{to } \lim_{m \rightarrow \infty} \mathbb{E}_{S \sim \mathcal{D}^m} \theta_m = 0, \text{ aka " } L^1 \text{ " convergence}$$

see Exercise 4.1

$$\theta_m \xrightarrow{L^p} \theta \text{ means}$$

$$\lim_{m \rightarrow \infty} \mathbb{E} |\theta_m - \theta|^p = 0$$

Other notions:

vs.

convergence in measure/probability

$$\theta_m \xrightarrow{P} \theta \text{ means } \forall \varepsilon > 0,$$

$$\text{Note } \theta_m \xrightarrow{L^1} \theta \Rightarrow \theta_m \xrightarrow{P} \theta$$

$$\lim_{m \rightarrow \infty} \underbrace{\left[\mathbb{P}[\{\omega : |\theta_m(\omega) - \theta(\omega)| > \varepsilon\}] \right]}_{\delta_m \text{ for us}} = 0$$

vs. almost sure convergence,

$$\mathbb{P}[\{\omega : \theta_m(\omega) \rightarrow \theta(\omega)\}] = 1 \quad \text{i.e. ptwise a.e.}$$